

GRAD-E1461: Agency and the Foundations of Technology Ethics**1. General information**

Course Format	Onsite
Instructor(s)	Joanna Bryson
Instructor's E-mail	Bryson@hertie-school.org
Assistant	Ayamba Kwoyila
Instructor's Office Hours	Vary by the week due to her frequent public and policy engagements. Link to online sign up available from Moodle and through the assistant.

2.

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course room, times and session dates**, please consult the [Course Schedule](#) on *MyStudies*.

Instructor Information:

Joanna Bryson is Professor of Ethics and Technology at the Hertie School. With broad academic expertise on intelligence, its nature, and its consequences, Bryson actively participates in technology policy making globally, but with particular foci on AI and the EU. Though educated primarily in the behavioural and computer sciences, Bryson's increasing engagement with policy makers and processes led her to the Hertie School, where she started 1 February 2020 (Brexit day). Five months later, in July 2020 she became one of Germany's nine appointees to the first cohort of experts for the G7 Global Partnership for AI. Previously, Bryson was Computer Science faculty at the University of Bath (2002-2019); she has also been affiliated with Harvard Psychology, Oxford Anthropology, Mannheim Social Science Research Centre, The Konrad Lorenz Institute for Evolution and Cognition Research, and the Princeton Center for Information Technology Policy. Bryson's PhD was on the systems engineering of AI, and she has long researched AI transparency (originally for AI development) as well as the biological utility and impacts of intelligence. During her PhD, she happened to observe confusion generated by anthropomorphised AI, leading to her first AI ethics publication "Just Another Artifact" in 1998. From this a thread of ethics and advocacy led to Bryson co-authoring the first national level AI ethics policy, the UK's 2010 Principles of Robotics, and from there to advising IEEE, OECD, Google, EU, and UN efforts (among others) on regulating AI. Bryson is a founding professor of Hertie's Centre for Digital Governance. Her present research interests include AI impacts on human cooperation and coordination, power and market impacts of transnational technology services and their regulation, and digital impacts on security, democracy, and human understanding, though she also speaks frequently in the context of philosophy and the humanities, particularly in the context of AI consciousness and human agency. Bryson holds two degrees each in psychology and AI (BA Chicago, MSc & MPhil Edinburgh, PhD MIT) and a *doctorats honoris causa* (Sciences and Architecture) from Université de Louvain.

3. Course Contents and Learning Objectives

Course contents:

Who or what has agency in our societies? What determines our individual policy preferences? To whom or to what are we most obliged? How can we ensure we are able carry out our obligations? Is it possible to have obligations that are impossible to address?

The foundations of all policy and governance are both behavioural and moral. Governance and policy concern fundamental beliefs about what matters, as well as fundamental capacities to create security and a sense of legitimacy, allowing governance to function. These arise from absolute physical limits on what is not only doable but knowable, and also on tendencies of humanity as a species – capacities and perhaps inclinations we may alter when we innovate and interact with technology. Human motivations and abilities are often shared with other biological species and even non-living complex systems. But moral beliefs are generally considered to be cultural, and have until recently been unquestionably held only by humans. Even within humans, different ethical systems not only recommend different goals, but even start from different assumptions about what is essential.

This course examines what it means to be an agent, and further, what it means to be a *moral* agent – an agent responsible for its actions. We examine both (1) theories or traditions of ethics, and (2) the scientific undergirdings of individual and social agency. We look broadly at technology history, science, and ethics to discover precedents to and analogues of contemporary problems with agentic artefactual systems. We also examine how and when ethics impacts our behaviour, ranging from the implicit action choice through political debates, and how various understandings of morality help individuals and collectives achieve their goals. Topics include consciousness, evolution, sociality, identity, ingroup-outgroup dynamics, (affective) political polarisation, cooperative and competitive behaviour, inequality and extremism, historic interpretations of ethics, and the inclusion of AI in systems of responsibility. Policy and governance stand as a background motivation to the course, but are not the primary topics.

Note: this course does not cover some common topics in "AI Ethics" such as bias, surveillance, or military use that are already well-developed in policy and law – for these see GRAD-E1319: Governance and Politics of Artificial Intelligence

Students will be marked through examination, presentations, and optionally on a short scientific paper using AI models (agent-based modelling, ABM) to make theoretical claims. ABM is used throughout the course to ensure students have direct experience of building AI, and students should bring laptops to lecture with NetLogo preloaded. ABM are used in order to develop stronger intuitions and understanding of intelligent behaviour and its collective, ecological, systemic outcomes. ABM is for example useful for understanding the aggregate consequences of working cooperatively or competitively to produce and exploit public goods. Students are also offered but not mandated the opportunity to assume hands-on experience of deploying and even developing such models in policy contexts, (the alternative is a second, final exam).

Learning objectives:

By the end of this course, students should be conversant in the basics of both moral philosophy and the sciences of intelligence, with a particular focus on intelligent technology. They should be able to both introduce and utilise terminology from these fields to argue coherently with reference to a current policy goal and political context. This includes being able to take as a starting point a variety of different moral framings – a useful skill particularly for international policy debates, and embracing the opportunities of a growing, developing, and increasingly mobile world. Students should understand the problems of generating appropriate action in a context, how this problem is handled by individual intelligences, and the ways in which systems such as ethics, policy, commerce, governments, and indeed prompts alter the landscape of what intelligence can do, as well as what it is likely to do. They should also be introduced to current concerns of technology policy, such as responsibility gaps and the

moral status of AI.

Target group:

Second year MIA, MPP, or MDS students interested in governing humans, and engaging in policy debate and negotiations with less-conventional political actors such as technology enthusiasts or religious organisations. Some material may also be useful for understanding broader concerns such as sustainability, transhumanism, and the potential for fully artificial societies, but these are not the primary focus, and questions to these ends will never be mandatory on exams (though may be available). While we will discuss in depth human rights or sustainability, we will be working to understand the ethics and science behind both human and ecological flourishing. This course should be of particular interest to those concerned with technology policy, including for intelligent technology.

Teaching style:

This course is very dependent on the readings having been done in advance, including in some cases reading of code and models. The models are in NetLogo, a language designed for teaching programming to children, consisting largely of ordinary English, but of course with some logic. There are three main types of lectures:

1. Lectures primarily given by the professor to complement (extend) material available in the readings. 2 lectures.
2. Lectures with an hour devoted to student-led discussion and debate of reading and lecture material (students will be marked for this.) 4 lectures.
3. Lectures with at least a half-hour devoted to student-led discussions of ABM code and simulation outcomes. 4 lectures (the second half hour is optionally ABM or readings discussion.)

There is also one lecture devoted entirely to a guest lecture, in congruence with an in-class midterm exam, and one to presentations of student modelling projects and/or further guest speakers. All students are also required to keep up with Moodle announcements and encouraged to participate on Moodle forums.

Prerequisites:

None, but students must be willing and able to discuss and critique ethical systems that may be at variance with their own closely-held ones. Students should be able to accept for the sake of an argument, or perhaps over the course of an assignment, a set of principles that may not agree with their own, and be able to demonstrate that they can reason within that framework and show some of its consequences. Students must also be willing to really read the textbook.

Diversity Statement:

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness are integrated in the course content whenever possible. Most of the required readings are from two books by ethnic Europeans, but both are informed by a far broader literature. Guest speakers are chosen to further extend philosophical perspectives. Students are encouraged to bring in further literatures, views, and life experiences to inform all discussions. Course direction is informed by the professor's real-world experience of policy debates in transnational, including global contexts, and particularly by efforts to communicate policy legitimacy to a large general public.

3. Grading and Assignments

Assignments are extremely flexible in some respects. For example, students may choose to present

once by themselves or twice with partners, provided they can find partners who agree to work with them. Similarly, students have the choice of whether to do a second, final exam, or to do a short project using ABM to explore policy-relevant options to present technology policy concerns. Time will be set aside during the first lecture to allow students to select weeks and form pairs or (depending on numbers) groups (where desired) for their presentations. The decision about modelling can be made up until the 10th lecture, which means it can be decided after receiving feedback on the midterm exam.

Policy on the use of AI tools for assignments

AI guidance specific to this class is in the section **Academic Integrity** below. The boiler plate advise still holds: Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

Composition of Final Grade:

1 or 2 In-Class Discussions	Determined during first week.	In agreed lecture. Slots can be swapped with consent of all student parties involved in swap.	26% total
Midterm Exam	week of lecture 6 or 7		30%
ABM-Based Policy Project <i>or</i> Final Exam	ABM Deadline: 12th lecture. Report deadline: Midnight Friday before exam week.	ABM: Submit via Moodle	44%

Assignment Details

The aim of the course overall is to ensure that all students can recognise, understand, and communicate the value and use of ethical and scientific reasoning about agency, particularly in policy contexts. The primary value of the course to the students should be in the readings, lectures, discussion, and the

knowledge attained through these methods. However, it is also necessary to provide an assessment of how much knowledge is being internalised by each individual student, and their ability to utilise and communicate this knowledge. Because such capacities often need to be expressed in real-time policy negotiations, they are tested by in-class exam, and in-class discussions.

Assignment aims:

- All the assignments (even the exams) are designed to help you think critically about the information you are being presented. Note that critical thinking is not necessarily negative thinking. A good critic highlights the best of available work as part of the process of seeking constructive means to improve our collective understanding and situation. Note that the first time the course was offered, exam performance improved significantly for the majority of students between the two exams. Others were good from the start and/or chose the ABM assignment. (The second time it was offered, they all students chose the ABM. You are the third.)
- The optional policy project also offers an opportunity to creatively engage with and apply the concepts learned in this class, which helps you with understanding material more deeply.
- Finally, of course, the assessment gives you a chance to demonstrate your knowledge and skills in using the material of the course.

In-class discussions: Students should coordinate with any other presenters on the same day to ensure they are not working on the same topic, or if they are, that there is a good reason for it e.g. a very different perspective taken or outcome. Each discussion should begin with one or two students giving a **short, 5-6 minute speech or presentation, or 8-10 minutes if presenting a model**. A speech should make an interesting claim about the course material for that lecture. They have 5-6 minutes to do that, unless they have chosen to discuss a model, in which case they have. The rest of their 25 minute slot should consist of fielding discussion from the floor (the other students) concerning their presentation. Marks will be assigned not only for the quality of the presentation, but the ability to internalise any suggestions raised (positive or negative!) and come to a negotiated understanding. Discussions must be related to the specific lecture materials, not other favourite topics (e.g. not bias, not LAWS.)

Exams: Students will be confronted by three to five short, abstract questions of ethics and/or science. They will need to write an essay in which they first define their terms and goals, then make their arguments. They may answer any number of the questions, though at least two are advised. Marks are for the overall quality of their paper in terms of both argument and contact to the course materials and in-class discussion. **Note: the two exams are equally long, and equally difficult.** The first one is just worth fewer points because you may not have done a real-time written exam for a while. It is basically a "practice" exam, but for mark of course.

Policy project and presentation (optional): Rather than sitting a second exam, students may choose to work on an ABM either by themselves or with one other student. Students are to select a contemporary policy concern congruent with their interests and/or expertise, and then use an agent-based model to illustrate and explore key potential tradeoffs relevant to a policy choice. Students will prepare a short document (2-3 pages, one author; 4-5 pages for two) and an eight minute presentation. In both the paper and the presentation, students will present:

1. the policy context,
2. the model, which may be one written by the students, one presented in class (with, or without modification), or one discovered elsewhere (ditto). **All modifications should be clearly commented in code,**
3. one or two graphs illustrating and analysing model outcomes,
4. a description of empirical steps that could be taken as further research to check whether the predictions of the model are likely to be correct, and finally,

5. one or two policy recommendations, assuming empirical data supports the model.

In such a project, you will need to concisely summarise the current state of scientific knowledge of a specific aspect of a policy domain. Note that any policy recommendations or conjectured system dynamics have to be realistic and practical, taking into account everything you learnt at Hertie School (and beyond) about political and social behaviour, public policy, and policy making. Note too that there is no 'perfect project'. While you are encouraged to be ambitious, the most important aspect of such a research project is your learning experience. Hence, you don't want to pick something that is too easy for you, but similarly, you don't want to choose a project that might be out of the scope of this class, too difficult to communicate quickly and effectively, or too complex to model. The objective of the policy project is to provide you with hands-on practice and an opportunity to learn. Quite evidently, you don't want to "write" something you don't understand or haven't verified; cf. academic integrity guidance below.

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance: Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session.** For further information please consult the [Examination Rules](#) §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#). Students may use genAI responsibly to better inform themselves and to correct their English use and writing style, but should never use it as a substitute for their own understanding or creativity, including that it is not advised to use it for reading summary – learning to skim is a far more useful skill. See further [Generative AI Use and Human Agency](#), guidance written by the professor.

Compensation for Disadvantages: If a student furnishes evidence that they are not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline. Note that some of this course is group work, so contacting other students in your group can be at least as important as contacting the professor.

4. Course Sessions and Readings

The Required readings are really very required, in advance. The first lecture has a lot of readings, which might well be done before term starts. In contrast, the optional readings are really very optional, more like a bibliography in case you want to pursue a concept further, perhaps even after the lecture.

The primary text for this course is Sven Nyholm's "This Is Technology Ethics". There are also several draft chapters from the professor's own forthcoming book, but these are less essential. Please note that this is not a full endorsement of Nyholm's positions, but rather that the book is a good and thorough read from a professional moral philosopher. One of the considerations of this course is: in what ways and to what extent moral philosophy can and should influence policy, whether corporate, government, or personal.

Students may also find the following books helpful, which are all available from the library:

- Smaldino, Modeling Social Behavior: Mathematical and Agent-Based Models of Social Dynamics and Cultural Evolution
- Carlson, Physiology of Behaviour
- Boyd & Silk, How Humans Evolved.
- Mitchell: Artificial Intelligence: A Guide for Thinking Humans

Of these, probably Mitchell's is most helpful and the best prereading, **if you have not previously studied how AI really works.**

Students should also read the NetLogo user guide.

Note: the readings below are almost entirely inaccurate, from last year. I only updated the session titles. Note! But the prereadings above are right; read those.

Session 1 (model): Ethics, Agency, and Policy	
Required Readings	• Nyholm Chapters 1 and 2 • Bryson, Joanna J. "Human Experience and AI Regulation: What European Union Law Brings to Digital Technology Ethics." Weizenbaum Journal of the Digital Society 3, no. 3 (2023).
Optional Readings	• DRAFT Bryson, 10.000 years of AI, Chapter 1 • Chapter 2 "Intelligent Agents" of Russell & Norvig, AI. • Naomi Alderman, "Nick Clegg and Tim Berners-Lee: the battle for the soul of the internet", The Observer, Saturday 6 September 2025. URL • Bryson, Joanna J. "Robots should be slaves." In Close engagements with artificial companions: Key social, psychological, ethical and design issues, pp. 63-74. John Benjamins Publishing Company, 2010.URL • Bryson, Joanna J. "Robots should be slaves." In Close engagements with artificial companions: Key social, psychological, ethical and design issues, pp. 63-74. John Benjamins Publishing Company, 2010.

Session 2: Intelligence, Brains, and Patiency: Intro to Free Will

Required Readings	• Nyholm Section 3.1 (pp. 51-53), on "Reflective Equilibrium" as a method for ethics. • Nyholm Chapters 7-8 • Braitenberg, Valentino. Vehicles: Experiments in synthetic psychology. MIT press, 1986. (in library, particularly Ch 1-4, pp. 1-15) > In the Library in reserve collection (l.02.06_46). • Bryson, Joanna J. "Patency is not a virtue: the design of intelligent systems and systems of ethics." Ethics and Information Technology 20, no. 1 (2018): 15-26.
Optional Readings	Strongly advised: • Brooks, Rodney A. "Intelligence without representation." Artificial intelligence 47, no. 1-3 (1991): 139-159. • Wall Following Model in NetLogo Normal level of optional: • Chapter 6, Vision, Carlson & Birkett, Physiology of Behaviour • Chapter 7, Audition, the Body Senses, and the Chemical Senses, Carlson & Birkett, Physiology of Behaviour • Brooks, R. "Intelligence without Reason. IJCAI'91, 12th Int." In Joint Conference on Artificial Intelligence, Sydney, Australia. 1991. • David Cassel, "The Thorny Ethics of Computer Programming," The NewStack Dec 11th, 2016 7:28am Very optional • Durstewitz, Daniel, Nicole M. Vittoz, Stan B. Floresco, and Jeremy K. Seamans. "Abrupt transitions between prefrontal neural ensemble states accompany behavioral transitions during rule learning." Neuron 66, no. 3 (2010): 438-448. 10 • Johannes Vermeer Girl with a Pearl Earring

Required Readings	• Bryson DRAFT Chapter 2 Computability, models & bias • Hinton & Nolan, How learning can guide evolution • Turing Machines, in the Stanford Encyclopedia of Philosophy
Optional Readings	• Bayes' Rule – Explained For Beginners, from free code camp • McClelland, Tom. "The mental affordance hypothesis." <i>Mind</i> 129, no. 514 (2020): 401-427. • Chapter 1, How humans evolved, Boyd & Silk (available from the library) Section 4.1.4 on genetic algorithms, Third edition • Chapter 3&4, How humans evolved, Boyd & Silk (available from the library) • Chapter 13, Learning and Memory, Carlson & Birkett, <i>Physiology of Behaviour</i> at least skim for comparison to R&N • Section 4.1.4 on genetic algorithms, Third edition (will add section for Fourth once I've looked at it, might differ). Also sections 1 (introduction) of all three learning chapters (18, 19, 20 in the Third edition) Russell & Norvig, AI. Very optional: • Campàs, Otger, Ricardo Mallarino, A. Herrel, Arkhat Abzhanov, and M. P. Brenner. "Scaling and shear transformations capture beak shape variation in Darwin's finches." <i>Proceedings of the National Academy of Sciences</i> 107, no. 8 (2010): 3356-3360. • Hinton, G. E.; Osindero, S.; Teh, Y. W. (2006). "A Fast Learning Algorithm for Deep Belief Nets" <i>Neural Computation</i>. 18 (7): 1527–1554. doi:10.1162/neco.2006.18.7.1527.

Session 4 (model): Models, Abstraction, Science, Awareness, Segregation

Required Readings	• Thomas C. Schelling (1971) Dynamic models of segregation, Journal of Mathematical Sociology, 1:2, 143-186, DOI: 10.1080/0022250X.1971.9989794 • Schelling model in NetLogo • Hinton & Nolan, How learning can guide evolution • Bryson, Joanna J., Yasushi Ando, and Hagen Lehmann. "Agentbased modelling as scientific method: a case study analysing primate social behaviour." Philosophical Transactions of the Royal Society B: Biological Sciences 362, no. 1485 (2007): 1685-1699. Introduction (pp.1685-1687)
Optional Readings	• Planetary Motion: The History of an Idea That Launched the Scientific Revolution, https://earthobservatory.nasa.gov/features/OrbitsHistory • DRAFT Bryson, 10.000 years of AI, Chapter 3 (part b)

Session 5 (model): Emotions, Dominance, and Differential Learning Systems

Required Readings	• DomWorld replication model(s) • Bryson, Joanna J., Yasushi Ando, and Hagen Lehmann. "Agent-based modelling as scientific method: a case study analysing primate social behaviour." Book chapter version (after introduction!) • Nyholm, the rest of Chapter 3. • Hamann, Stephan. "Mapping discrete and dimensional emotions onto the brain: controversies and consensus." Trends in cognitive sciences 16, no. 9 (2012): 458-466.
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Optional Readings	• Other model shown briefly of dominance and gender • Thierry, Bernard. "Unity in diversity: lessons from macaque societies." <i>Evolutionary Anthropology: Issues, News, and Reviews</i> 16, no. 6 (2007): 224-238. • How humans evolved, Boyd & Silk, pp 136-155 (available from the library) • The rest of Chapters 5&6, How humans evolved, Boyd & Silk (available from the library) • Carlson & Birkett. <i>Physiology of Behaviour</i>, Chapter 11, Emotion, particularly pp. 365-379, Very optional • Rohlfshagen, Philipp, and Joanna J. Bryson. "Flexible latching: A biologically-inspired mechanism for improving the management of homeostatic goals." <i>Cognitive Computation</i> 2, no. 3 (2010): 230- 241. • "How to make a monkey do something smart", Ch 2 of Bryson,J.J., 2001. <i>Intelligence by design: principles of modularity and coordination for engineering complex adaptive agents</i> (Doctoral dissertation, Massachusetts Institute of Technology). • Bann, Eugene Y., and Joanna J. Bryson. "The conceptualisation of emotion qualia: Semantic clustering of emotional tweets." In <i>Computational models of cognitive processes: Proc 13th neural computation & psychology workshop</i>, pp. 249-263. 2014. • Eugene Y. Bann & Joanna J. Bryson "Measuring Cultural Relativity of Emotional Valence and Arousal using Semantic Clustering and Twitter", <i>Proc. 35th Conference of the Cognitive Science Society (CogSci)</i>, Berlin, August 2013, pp. 1809–1815.
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Session 6 (long lecture): Language, Sentience, Consciousness, and Morality	
Required Readings	• Čače, Ivana, and Joanna J. Bryson. "Agent based modelling of communication costs: Why information can be free." In <i>Emergence of communication and language</i>, pp. 305-321. Springer, London, 2007. • Čače and Bryson NetLogo code (revised)

Optional Readings	• Chapters 8 & 16, How humans evolved, Boyd & Silk (available from the library) • McClelland, J L, B L McNaughton, and R C O'Reilly. "Why there are complementary learning systems in the hippocampus and neocortex: insights from the successes and failures of connectionist models of learning and memory." Psych. review 102(3):419 (1995). File • Supervisory attentional system – Norman & Shallice 1980, 1986 (Wikipedia) • Cultural Evolution in the Digital Age, by Alberto Acerbi • Susanne Shultz and Christopher Opie and Quentin D. Atkinson, "Stepwise evolution of stable sociality in primates, Nature, 2011. • Powell, Adam, Stephen Shennan, and Mark G. Thomas. "Late Pleistocene demography and the appearance of modern human behavior." Science 324, no. 5932 (2009): 1298-1301.
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Session 7: Midterm and Guest Speaker

Required Readings	• (may change) Paper by Muindi Fanuel - "AI&D: Signal-Processing, Noise-Making, and the Right to Opacity"
Optional Readings	

Session 8 (model): Public Goods , Cooperation, and Identity

Required Readings	• Dennett, Daniel C. From bacteria to Bach and back: The evolution of minds. Chapter 15. WW Norton & Company, 2017. • Bryson, Joanna J. "Embodiment versus memetics." Mind & Society 7, no. 1 (2008): 77-94. • McClelland, Tom. "The mental affordance hypothesis." Mind 129, no. 514 (2020): 401-427. File
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Optional Readings	• Caliskan, Aylin, Joanna J. Bryson, and Arvind Narayanan. "Semantics derived automatically from language corpora contain human-like biases." <i>Science</i> 356, no. 6334 (2017): 183-186. • McClelland, J L, F Hill, M Rudolph, J Baldridge, and H Schütze. "Placing language in an integrated understanding system: Next steps toward human-level performance in neural language models." <i>Proc National Acad of Sciences</i> 117, no. 42 (2020): 25966-25974. • Watkins, CD, D Xiao, & DI Perrett. "Social transmission of leadership preference:...group membership & partisan media reporting moderates perceptions of leadership ability from facial cues to competence & dominance." <i>Frontiers in Psychology</i> 10 (2020): 2996 • Bryson, Joanna J. "A role for consciousness in action selection." <i>International Journal of Machine Consciousness</i> 4, no. 02 (2012): 471-482. Very optional: • Floresco, Stan B., Jeremy K. Seamans, and Anthony G. Phillips. "Selective roles for hippocampal, prefrontal cortical, and ventral striatal circuits in radial-arm maze tasks with or without a delay." <i>Journal of Neuroscience</i> 17, no. 5 (1997): 1880-1890. • Baars, B. J., Franklin, S., & Ramsøy, T. Z. (2013). Global workspace dynamics: cortical "binding and propagation" enables conscious contents. <i>Frontiers in psychology</i>, 4, 200. • Franklin, Stan, Steve Strain, Ryan McCall, and Bernard Baars. "Conceptual commitments of the LIDA model of cognition." <i>Journal of Artificial General Intelligence</i> 4, no. 2 (2013): 1-22. • Powell, Adam, Stephen Shennan, and Mark G. Thomas. "Late Pleistocene demography and the appearance of modern human behavior." <i>Science</i> 324, no. 5932 (2009): 1298-1301.
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Required Readings	- Herrmann, Benedikt, Christian Thöni, and Simon Gächter. "Antisocial punishment across societies." <i>Science</i> 319, no. 5868 (2008): 1362-1367. - Polarization under rising inequality and economic decline Alexander J. Stewart, Nolan McCarty, Joanna J. Bryson 2020 - Polarisation Netlogo Code (Max Hampel)
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Optional Readings	- Documentation for polarisation code - Herrmann &al. 2008 Supplement. - Stewart &al. supporting info - Krosch, Amy R., and David M. Amodio. "Economic scarcity alters the perception of race." <i>Proceedings of the National Academy of Sciences</i> 111, no. 25 (2014): 9079-9084. - Lai, Calvin K., Allison L. Skinner, Erin Cooley, Sohad Murrar, Markus Brauer, Thierry Devos, &al. "Reducing implicit racial preferences: II. Intervention effectiveness across time." <i>Journal of Experimental Psychology: General</i> 145, no. 8 (2016): 1001. - Wasow, Omar."Agenda Seeding: How 1960sBlack Protests Moved Elites, Public Opinion and Voting." <i>American Political Science Review</i> (2020): 1-22. - Reynolds-Stenson, Heidi. "Protesting the police: anti-police brutality claims as a predictor of police repression of protest." <i>Social movement studies</i> 17, no. 1 (2018): 48-63. K. Sylwester, J. Mitchell, W. Lowe and J. J. Bryson (draft) "Uses and consequences of costly punishment across populations" - Bryson: DRAFT Chapter 3, pages 80-104 (see lecture 3 for link)
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Session 10: Responsibility, Praise, and Blame	
Required Readings	- Nyholm Chapters 4-6 - Kneer, Markus, and Markus Christen. "Responsibility gaps and retributive dispositions: Evidence from the US, Japan and Germany." draft near publication in <i>Science and Engineering Ethics</i>, 2024

Optional Readings	• Leib, M., Köbis, N., Rilke, R. M., Hagens, M., & Irlenbusch B. (2024). Corrupted by algorithms? How AI-generated and human-written advice shape (dis)honesty. <i>Economic Journal</i>, 134(658), 766–784. URL • Stuart, Michael T., and Markus Kneer. "Guilty artificial minds: Folk attributions of mens rea and culpability to artificially intelligent agents." <i>Proceedings of the ACM on Human-Computer Interaction</i> 5, no. CSCW2 (2021): 1-27. • Vernor Vinge, "The Coming Technological Singularity: How to Survive in the Post-Human Era", 1993 (webpage / rant) • Elhacham, E., Ben-Uri, L., Grozovski, J. et al. Global human-made mass exceeds all living biomass. <i>Nature</i> 588, 442–444 (2020)
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Session 11: Friendship, Virtue, and Machines	
Required Readings	• Nyholm Chapters 9 & 10; • Vallor, Shannon. "Flourishing on facebook: virtue friendship & new social media." <i>Ethics and Information technology</i> 14 (2012): 185-199. • Wang, G., Wang, W., Cao, Y. et al. Possibilities and challenges in the moral growth of large language models: a philosophical perspective. <i>Ethics Inf Technol</i> 27, 9 (2025).

Optional Readings	• UNESCO AI Ethics Recommendation (2022) • Evans, Katie D., Scott A. Robbins, and Joanna J. Bryson. "Do we collaborate with what we design?." <i>Topics in Cognitive Science</i> (2023) • "We only learnt of our son's secret online life after he died at 20", by Ben Machell Saturday October 05 2024, <i>The Times</i> • Van Wynsberghe, Aimee, and Scott Robbins. "Critiquing the reasons for making artificial moral agents." <i>Science and engineering ethics</i> 25, no. 3 (2019): 719-735. • Floridi - Hyperhistory, the emergence of the MASs, and the design of infraethics.
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Session 12: Students and/or Guests

Required Readings	
Optional Readings	• Tips on communicating scientific results to policy makers: https://www.forbes.com/sites/marshallshepherd/2016/11/22/9-tipsfor-communicating-science-to-people-who-are-notscientists/ (May change) Forthcoming in Ted Peters (ed.) in 'AI and IA: Utopia or Extinction?' Agathon (Hindmarsh, ATF) Between Fear and Hope: Some notes on AI ethics and the potential of Islamic Thought, by Ali-Reza Bhojani